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Don't Fall for the Siren-Song Price of Under-powered Desktop PCs

IS executives have been captivated by the siren song of sub-\$1,000 PCs, thinking they could save millions on corporate desktop-PC buying. And why not? Buyers are seeing twice the CPU clock speed and twice the memory of a couple of years ago at bargain basement prices. But recent Aberdeen research indicates many present and almost all future corporate desktops are better suited to more, not less.

Is Less Best?

The economics are compelling and simple; if astute IS managers can mandate less costly standard personal computer platforms, then they can easily save their enterprises millions of dollars a year at Global 1000-size organizations. The kind of decision that results in fat bonuses. As a result, many organizations in 1998 have been lagging behind Intel's new product introductions that are now up to 450MHz in speed in favor of less powerful platforms. Some are even buying home computers that sell for around \$1,000 in volume thinking "these seem to do the job OK." Until now, Aberdeen had little evidence to argue the fallacy of less is better.

What A Corporate Desktop Does Is Changing

Three years ago, a new desktop PC would typically run at 150 MHz and have a generous 32MB of main memory. Enough to run Windows 95, Office 95 and perhaps an e-mail client and file-sharing over a 10 Mbps LAN. The PC cost in volume was over \$2,000 per unit. Performance was quite acceptable with 32MB of memory but barely adequate with 16 megabytes of memory.

Fast forward to late 1998. The basic corporate desktop's workload looks on the surface about the same as three years ago. But research at Aberdeen Group shows a considerable increase in the loads placed on our own desktops. What was a superb PC a year ago — say a 233 MHz Pentium II with 32 MB — is becoming just adequate, and everything older is struggling — sometimes badly.

Today's desktops at Aberdeen Group are typically running:

- **Operating System:** Windows 95 OSR2 or Windows 98;
- **Net:** TCP/IP and NetBEUI;
- **Suite:** Office 97;
- **Browser:** Internet Explorer 4.0;

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- **E-Mail/PIM:** Outlook 98;
 - **Systems Management:** SNMP agent;
 - **Other:** anti-virus software, fat-client workflow applications, Intranet apps.

While the applications have often stayed the same, the versions have changed and are now consuming much more PC horsepower than ever before.

Where Are The MIPS Going?

Sure, we can blame feature-bloated versions of software. But the reality is that there has also been an explosion in services expected by desktop users and by IS managers:

- Intranet, Internet and Extranet support including background updates (e.g., channels and "push");
- A tidal-wave of electronic mail;
- Numerous shared-access network connections to workgroup data and applications;
- Desktop systems management capability to improve remote access and help desk diagnosis.

A lot of these MIPS-consuming services are occurring in background tasks invisible to the user. Moreover, the tasks take lots of main memory — the virtual memory usage on my own PC is about 60 MB — so 32 MB of memory is now too little.

Aberdeen's "Welded Case" Model For PC Buying

Aberdeen's own PC planning demands a 3-5 year asset life. Like most IS buyers, we hate the cost and aggravation of hardware upgrades because it means the labor-intensive opening of the desktop's case to install more memory or a processor upgrade. Multiplied by lots of desktops and it's problematic. We try to buy more PC power than is required at purchase knowing that software upgrades will inevitably consume the excess over the useful life of the machine.

What Is The Best Choice For The Future?

Like many IS executives, as CIO I had hoped to save money for the company by buying Celeron-class PCs or 233 MHz Pentium II inventory close-outs.

But our internal research, discussions with IT clients, and a view towards the near future convinced us that such a decision would soon prove to be a false economy. The Microsoft-Intel 1999 PC Standard specification calls for a minimum of a 300 MHz CPU and 32 MB of memory. Experience has shown that the minimum leads to less than satisfactory real-world performance that rapidly deteriorates over time — have you tried a 486 PC with today's software lately?

Ignoring the self-serving marketing hype from Intel touting high-end corporate desktops, Aberdeen concludes that buying up the PC food chain makes much more sense than buying down based on: maximizing the useful economic life of the PC; at

the minimum cost; with the highest performance levels. Spending more now will pay off across the PC's entire life cycle.

To buy a PC for knowledge workers in late 1998 for a truly useful life of at least three years, we recommend a 400 MHz Pentium II with 128 MB of memory and a 17" monitor. This configuration will support Office 2000 and Internet Explorer 5.0 in 1999 and an upgrade to Windows NT 5.0 in 2000 — while costing between \$1,800 and \$2,000 today, and less in three months.

— *Peter S. Kastner*

2024 Metadata

1. Concise Summary of the Findings

The document from Aberdeen Group, dated September 10, 1998, argues against the trend of purchasing sub-\$1,000 desktop PCs for corporate use. It highlights that while these cheaper PCs may seem cost-effective initially, they are underpowered for the increasing demands of modern software and network services. The report recommends investing in more powerful PCs to ensure longevity and better performance, suggesting a 400 MHz Pentium II with 128 MB of memory and a 17" monitor as the optimal configuration for corporate desktops.

2. Summary of How Computer Technology Buyers Should React

Computer technology buyers should avoid the temptation of purchasing low-cost, underpowered desktop PCs. Instead, they should invest in higher-end models that can handle the increasing demands of modern software and network services. This approach will maximize the useful economic life of the PCs, reduce the need for frequent upgrades, and ultimately save money in the long run.

3. Analysis of the Technology Discussed and Its Relation to Mainstream Computing in 2024

The technology discussed in the 1998 document, such as the 400 MHz Pentium II processor and 128 MB of memory, was considered high-end at the time. In 2024, mainstream computing has advanced significantly, with multi-core processors, gigabytes of RAM, and solid-state drives being standard. The principle of investing in more powerful hardware to future-proof against software demands remains relevant. Modern enterprises continue to prioritize robust hardware to support increasingly complex applications and services, including AI, big data analytics, and cloud computing.

4. Analysis of the Document's Impact on the History of the Companies Mentioned

The document's recommendations likely influenced corporate IT purchasing strategies, steering companies away from short-term cost savings towards long-term investment in more capable hardware. This shift would have benefited hardware manufacturers like Intel, which produced the recommended Pentium II processors, by driving demand for higher-end products. It also reinforced the importance of robust IT infrastructure in supporting business operations, a principle that continues to shape corporate IT strategies today.

5. Market Acceleration Catalysts

- Increasing software complexity and resource demands
- The need for longer hardware lifecycle and reduced upgrade frequency
- Rising importance of network services and internet connectivity
- Growing reliance on enterprise applications and systems management tools
- The push for higher productivity and efficiency in corporate environments

6. Keywords

corporate desktops, sub-\$1,000 PCs, Pentium II, 400 MHz, 128 MB memory, future-proofing, IT infrastructure, software demands, hardware lifecycle, enterprise applications

Citations:

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